Hamad Hassan

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Portfolio: hamadhassan-portfolio.github.io

PROFESSIONAL SUMMARY

Motivated Mechanical Engineering graduate with a strong grasp of industrial maintenance, reliability engineering, and continuous improvement methodologies. Proficient in leveraging MS Excel for data analysis, KPI reporting, and developing solutions for equipment performance optimization and downtime reduction. Eager to contribute to a dynamic maintenance team by applying analytical problem-solving skills and a commitment to operational excellence.

EDUCATION

Bachelor of Science in Mechanical Engineering — Expected 2027
University of Engineering and Technology, Taxila — *GPA: 3.1 / 4.0*

PROFESSIONAL EXPERIENCE

Heavy Mechanical Complex (HMC) Taxila — Jun 2026 – Present
Mechanical Engineering Intern — Taxila, Pakistan

- Assisted senior engineers in conducting visual weld quality inspections and monitoring thickness tolerances, ensuring adherence to quality standards.
- Monitored and documented equipment downtime logs, supporting maintenance teams in calculating Overall Equipment Effectiveness (OEE), Mean Time Between Failures (MTBF), and Mean Time To Repair (MTTR).
- Assisted in conducting 5S layout audits and workshop hazard mapping within machining bays to enhance safety and operational efficiency.

SELECTED ENGINEERING & INDUSTRIAL PROJECTS

Downtime Analysis and MTBF/MTTR Reliability Dashboard *Tools: MS Excel, Data Analysis, Pareto Principle*

- Designed an MS Excel KPI dashboard to analyze machine downtime, categorizing failures and calculating Mean Time Between Failures (MTBF) and Mean Time To Repair (MTTR) for a simulated manufacturing line.
- Modeled a predictive maintenance strategy that projected an 18% reduction in unplanned downtime by optimizing inspection intervals and addressing top failure modes.

Production Shop Floor Productivity and OEE Dashboard *Tools: MS Excel, Lean Manufacturing, OEE Metrics*

- Developed an Excel-based operational dashboard to monitor shop floor productivity and calculate Overall Equipment Effectiveness (OEE) by tracking availability, performance, and quality.
- Analyzed simulated production data to identify speed losses and bottlenecks, providing data-driven insights that projected an OEE improvement from 62% to approximately 74%.

TPM & 5S Shop Floor Re-layout for HMC Machining Workshop *Tools: AutoCAD, MS Excel, 5S Methodology, TPM Principles*

- Designed a Lean factory layout optimization plan for a heavy machining workshop, conducting safety audits and tool retrieval time studies.
- Redesigned tool storage and workstation placements using Total Productive Maintenance (TPM) and 5S principles, projecting a reduction in tool retrieval times from 4.5 minutes to about 45 seconds.

TECHNICAL & PROFESSIONAL SKILLS

- **Industrial Engineering**: Preventive Maintenance, Predictive Maintenance, Corrective Maintenance, Reliability Engineering (Mean Time Between Failures (MTBF), Mean Time To Repair (MTTR), Overall Equipment Effectiveness (OEE)), Rotating Equipment (Pumps, Motors, Compressors, Gearboxes), Hydraulic Systems, Pneumatic Systems, Bearings, Shafts, Couplings, Power Transmission, 5S Methodology, Total Productive Maintenance (TPM), Non-Destructive Testing (NDT), Boiler & Steam Systems, Compressed Air Audits, Centrifugal Pumps, Shaft Alignment
- **Software & Tools**: MS Excel (Calculations, Dashboards, VBA), MS Project, AutoCAD, SolidWorks, ANSYS
- **Standards & Compliance**: ASME Section VIII, ASME Sec V (Non-Destructive Testing), API 570

PROFESSIONAL CERTIFICATIONS

- Microsoft Office Specialist: Excel Advanced (Microsoft)
- Lean Six Sigma Green Belt
- Primavera P6 Professional Project Management Certification